

Introduction to Partial Differential Equations

Course:

Date:

Lecturer:

1. What is a PDE?

A **partial differential equation (PDE)** involves:

- Multiple independent variables (e.g. space and time)
- Partial derivatives

General form:

$$F\left(x, t, u, \frac{\partial u}{\partial x}, \frac{\partial u}{\partial t}, \dots\right) = 0$$

Where:

- x = space
- t = time
- $u(x, t)$ = unknown function

2. 1D Linear Convection Equation

$$\frac{\partial u}{\partial t} + c \frac{\partial u}{\partial x} = 0$$

3. Exact Solution

Given:

$$u(x, 0) = f(x)$$

Solution:

$$u(x, t) = f(x - ct)$$

4. Characteristics

$$x - ct = \text{constant}$$

5. Numerical Approximation

$$u_i^{n+1} = u_i^n - c \frac{\Delta t}{\Delta x} (u_i^n - u_{i-1}^n)$$

6. Stability (CFL)

$$\frac{c\Delta t}{\Delta x} \leq 1$$

Worked Example

Problem

Given the PDE:

$$\frac{\partial u}{\partial t} + 2\frac{\partial u}{\partial x} = 0$$

with initial condition:

$$u(x, 0) = \sin(x)$$

Step 1: Identify velocity

From the PDE:

$$c = 2$$

Step 2: Use general solution

$$u(x, t) = f(x - ct)$$

Step 3: Substitute initial function

Since $f(x) = \sin(x)$:

$$u(x, t) = \sin(x - 2t)$$

Final Answer

$$u(x, t) = \sin(x - 2t)$$

Interpretation

- The wave moves to the **right** at speed 2
- The shape (sine wave) does **not change**
- At time t , the wave is shifted by $2t$

Quick Check

At $t = 0$:

$$u(x, 0) = \sin(x)$$

✓ Matches initial condition

Summary

- Identify c from PDE
- Use $u(x, t) = f(x - ct)$
- Substitute initial condition